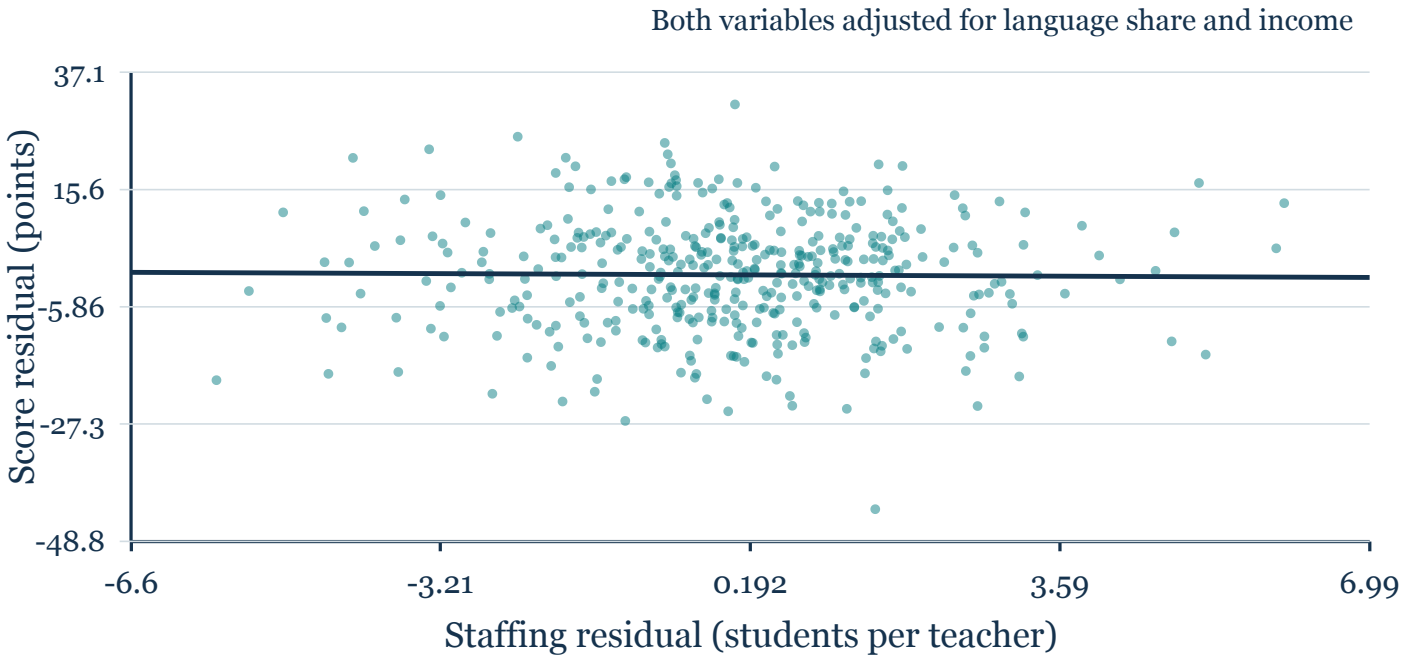
What does holding other things fixed mean?

Show the comparison left after observed controls are removed.



Actual residuals after removing English-learner share and income from both variables. The slope matches the full regression.

Recall why the short slope moved

A raw comparison can mix several observed differences.

> In the previous lesson, the short staffing slope combined the staffing relationship with an omitted language-composition contribution.

> Then identify what is left to understand.

The full model sets the target coefficient

Define every component before removing controls.

> $Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 W_i + u_i$

> $E(u_i \mid X_i, Z_i, W_i) = 0$ for the stated mean model

OLS solves a sample optimisation

Its residual identities follow from the fitted model.

> $\sum_i e_i = 0$

> $\sum_i X_i e_i = 0$ for an included regressor

Sample orthogonality is mechanical

Population exogeneity is a substantive assumption.

> Orthogonality means a zero inner product; here it is the zero sum of an included regressor times the OLS residual.

> Suppose recruitment responds to learning needs not captured by Z and W.

Residual versus error

Apply the idea before reading the solution.

Practice

OLS with an intercept gives residuals summing to zero. Does that verify $E(u \mid X) = 0$ in the population?

Residual versus error: explanation

Connect the calculation to the empirical claim.

> No. The zero residual sum is a normal-equation identity imposed by least squares. The unobserved population error and its relation to X are a substantive assumption. OLS can satisfy its sample identities in a badly confounded model.

First remove controls from the outcome

Save the score differences the controls do not predict.

> $rY_i = Y_i - \widehat{E}_{\text{linear}}(Y_i \mid Z_i, W_i)$

> $\widehat{E}_{\text{linear}}$: fitted linear prediction

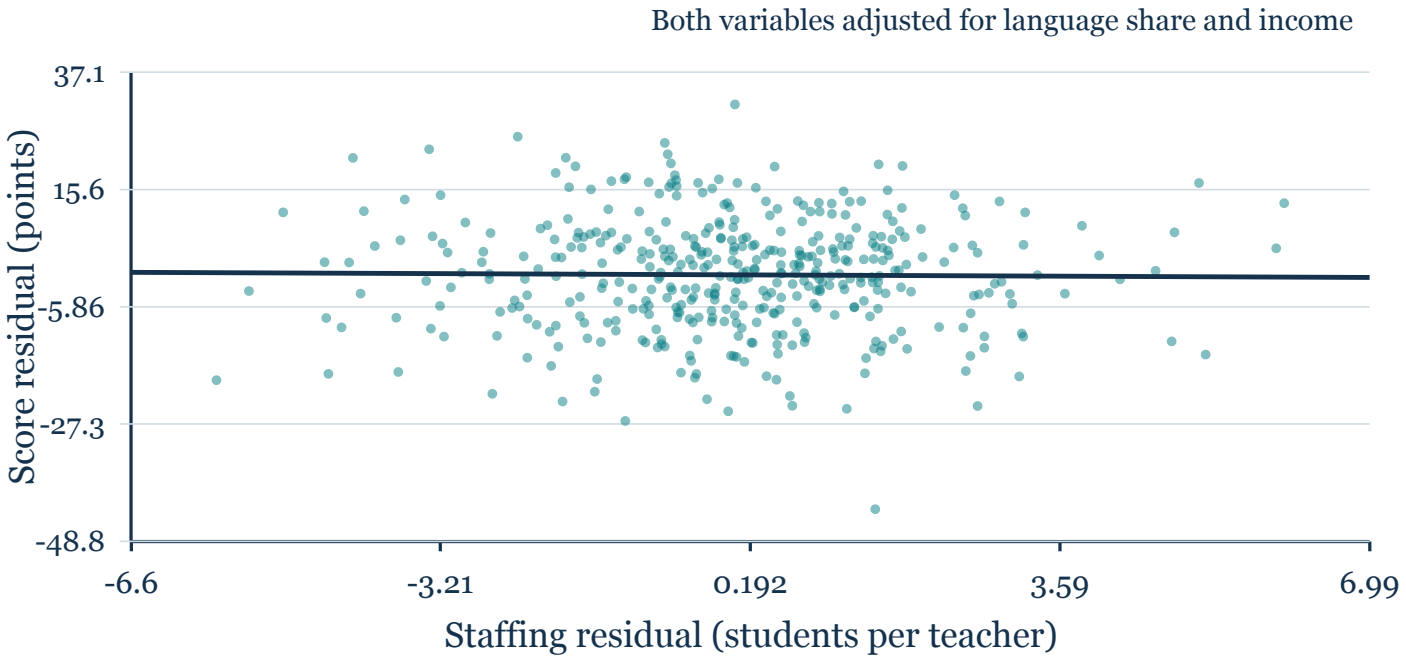
Then remove the same controls from staffing

An adjusted outcome must be compared with an adjusted regressor.

> $$rX_i = X_i - \widehat{E}_{\text{linear}}(X_i \mid Z_i, W_i)$$

Regress the two remainders on each other

The partialled slope equals the full-model staffing slope.



Actual residuals after removing English-learner share and income from both variables. The slope matches the full regression.

$$\hat{\beta}_1 = \frac{\sum_i rX_i rY_i}{\sum_i rX_i^2}$$

Read the full regression first

Keep its covariance calculation for inference.

```
. regress score str el income, vce(robust)

------------------------------------------------------------------------------
             |               Robust
       score | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         str |  -.0687755   .2901597    -0.24   0.813    -.6391376    .5015865
          el |  -.4882667   .0281161   -17.37   0.000     -.543534   -.4329995
      income |   1.494517     .09156    16.32   0.000     1.314539    1.674495
       _cons |   640.3155   5.991918   106.86   0.000     628.5373    652.0937
------------------------------------------------------------------------------
```

Verify the FWL identity in Stata

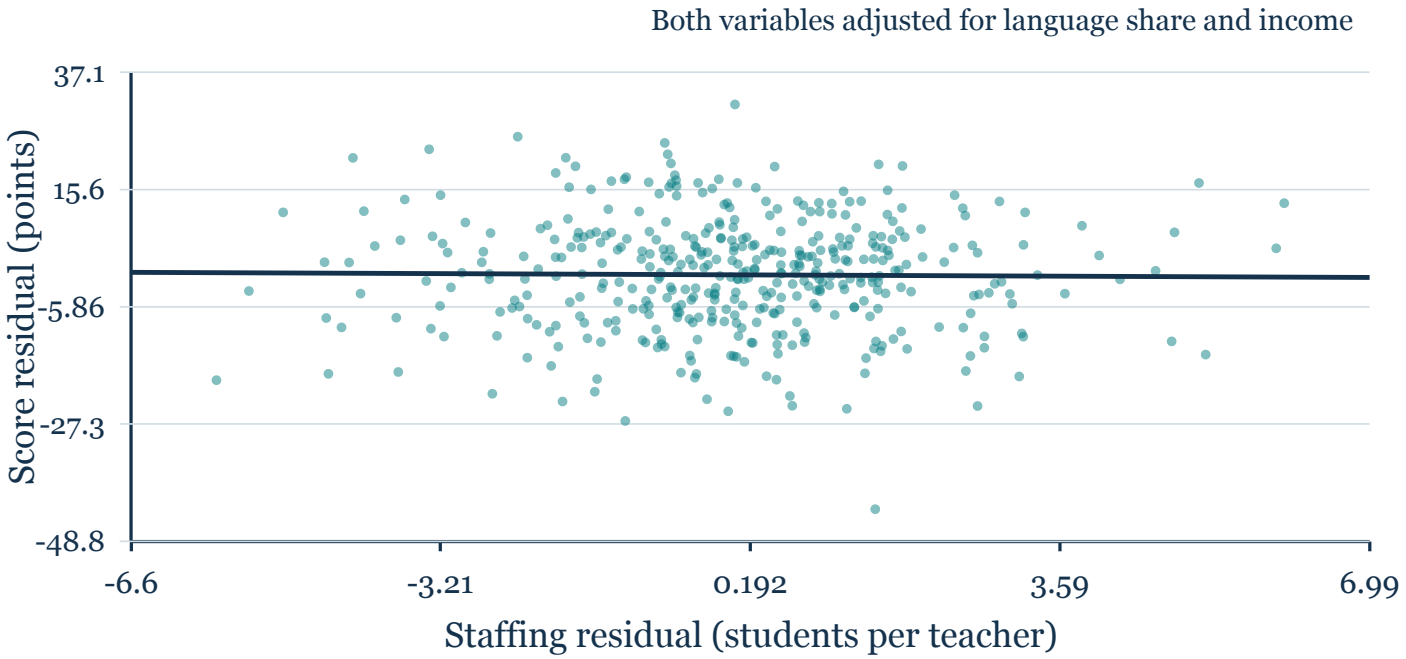
The slopes agree to numerical precision.

```
. regress ry rx, noconstant

------------------------------------------------------------------------------
          ry | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
          rx |  -.0687755   .2759147    -0.25   0.803    -.6111251     .473574
------------------------------------------------------------------------------
```

Read the residual axes in real units

They are deviations, not new measurements.



Actual residuals after removing English-learner share and income from both variables. The slope matches the full regression.

Partialling out

Apply the idea before reading the solution.

Practice

To estimate the schooling coefficient controlling for experience, describe the FWL procedure in words.

Partialling out: explanation

Connect the calculation to the empirical claim.

> Regress wages on an intercept and experience and save residuals. Regress schooling on an intercept and experience and save residuals. Regress the wage residual on the schooling residual (without an intercept). Its slope equals the schooling coefficient in the full regression. Standard-error reconstruction requires appropriate degrees of freedom and covariance treatment; the slope identity alone is the core result.

Unbiasedness concerns repeated samples

It does not promise an accurate estimate in every sample.

> $E(\widehat{\beta}_1) = \beta_1$ under the stated assumptions

Precision depends on usable regressor variation

Little variation after adjustment means a weak comparison.

> Usable variation: $\sum_i r X_i^2$

Multicollinearity reduces separate information

High correlation is different from exact dependence.

> Multicollinearity describes strong linear relationships among regressors.

> For example, including income measured in dollars and the same income divided by one thousand creates exact dependence.

Use VIF as a description, not a decision rule

It measures linear overlap among included regressors.

```
. quietly regress score str el income

. estat vif

    Variable |       VIF       1/VIF
-------------+----------------------
      income |      1.14    0.873929
          el |      1.12    0.891206
         str |      1.07    0.931159
-------------+----------------------
    Mean VIF |      1.11
```

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

R_j^2: fit from predicting regressor j with the other regressors

Efficiency needs additional error assumptions

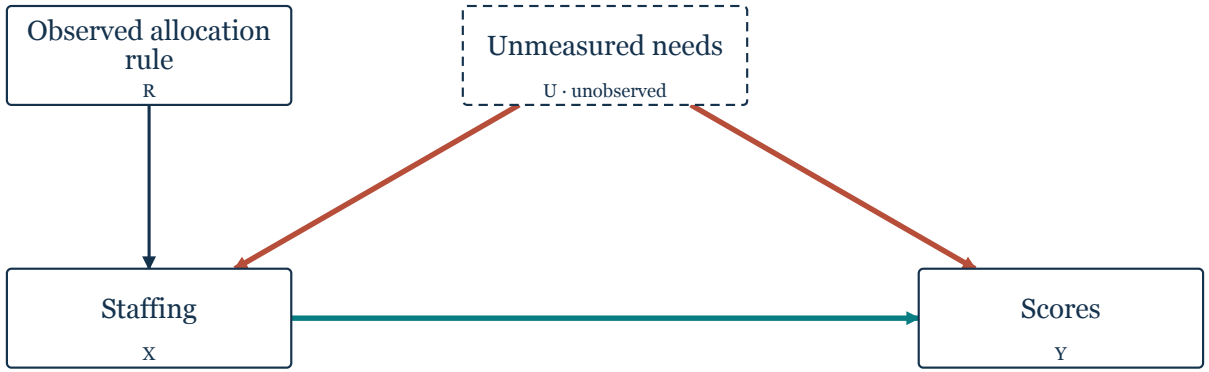
Small variance is a comparison among estimators.

> Efficiency means having less sampling variance than competing estimators within a stated comparison class.

> This does not mean OLS is always best for every task or every dataset.

A residual is not an automatically valid control

Its construction does not establish a causal role.



1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Collinearity

Apply the idea before reading the solution.

Practice

Two controls are highly correlated but neither is an exact linear combination of the others. Does this alone create omitted-variable bias?

Collinearity: explanation

Connect the calculation to the empirical claim.

> No. Under exogeneity, it need not bias OLS. It can reduce precision because there is little independent regressor variation. Exact collinearity is different: a separate coefficient is not identified in that specification.

Explain what the estimate uses

Describe the remaining comparison in ordinary language.

> For the school application, say that the staffing coefficient uses variation in student–teacher ratios left after their linear relationships with income and language composition are accounted for.

> Then state the remaining assumption: the adjusted staffing variation must not be related to relevant unobserved determinants in the way ruled out by the chosen error restriction.

Make the computation reproducible

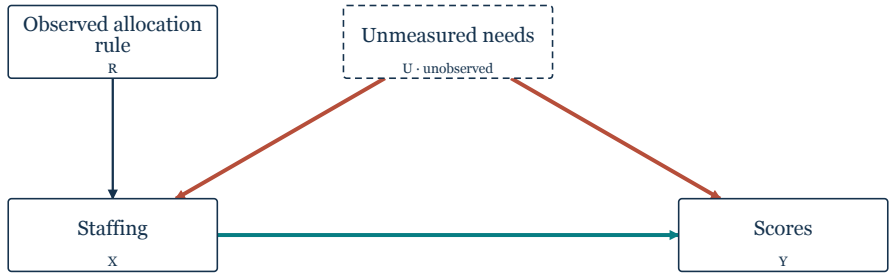
The same sample and control set are essential.

> The do-file fits the full model, saves the staffing coefficient, creates both supporting-regression residuals and checks the final slope.

> For reporting, retain the full model's chosen covariance estimate and the graph as an explanation of its identifying comparison.

DAG practice: partialling out and hidden needs

Trace the routes before choosing a control.



1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Practice

The observed allocation rule R helps explain staffing. Unmeasured needs U affect both staffing and scores. After residualising staffing and scores on R, has the graph-based identification problem necessarily disappeared? State the route and distinguish the calculation from the assumption.

DAG practice: partialling out and hidden needs: explanation

Explain what conditioning changes in this stated graph.

> The route $X \leftarrow U \rightarrow Y$ is still open. Removing the fitted contributions of R does not condition on the unobserved common cause U. The FWL coefficient identity can hold exactly while that identification problem remains. Residualisation describes which observed linear variation drives a coefficient. Exogeneity concerns whether the remaining comparison is unrelated to relevant omitted outcome determinants under the model. Those are different statements. A clear empirical report can verify the first by calculation while explaining why the second is, or is not, credible.

Verify partialling out without misusing its SE

Reproduce the evidence, then explain its economic meaning.

Practice

Run meeting_03.do. Describe what rx and ry measure, compare the residual-on-residual staffing slope with the full-model slope, and compare their printed SEs. Explain which result you would use to report robust uncertainty about the full staffing model and why.

Verify partialling out without misusing its SE: explanation

A reproducible result needs both a calculation and an interpretation.

> rx is staffing after removing its fitted relationship with the intercept, language share and income. ry is the score remainder after removing the same controls. The residual-on-residual slope is -0.068776, matching the full-model slope -0.068776. The manual residual regression prints SE 0.275915; the full regression's robust SE is 0.290160. These are different covariance calculations. FWL guarantees the coefficient identity for compatible fits, while inference must also account for the complete regression and covariance procedure. Use the full regression with its justified robust option when reporting that model's uncertainty. A residual plot is useful for understanding the adjusted comparison, but it does not establish exogeneity.

Next, measure uncertainty around the estimate

A valid calculation still needs an uncertainty statement.

> We have seen exactly how OLS constructs a coefficient and which sample identities it imposes.

> The next lesson asks how much an estimate would vary across comparable samples.